**before: u + v = 20 + 15 = 35**

**after: u + v = 25 + 10 = 35**

The assignment has not changed the value of u + v. We say that u + v is an invariant of the assignment. We can annotate before and after the assignment with the invariant expression.

**-- before: u + v = 35**

**u, v : = u + 5, v - 5**

**-- after : u + v = 35**

We can say, u + v is an invariant: it is 35 before and after. Or we can say u + v =35 is an invariant: it is true before and after.

**Example 8.2.** If we execute the following assignment with (p, c = 10, 9), after the assignment, (p, c) = (11, 10).

**-- before : p, c = 10 , 9**

**p, c := p + 1, c+1**

**-- after: p, c = 11 , 10**

Can you discover an invariant? What is the value of p - c before and after?

**before: p — c = 10 — 9 = 1**

**after: p — c = 11 — 10 = 1**

We find that p - c = 1 is an invariant.

In general, if an expression of the variables has the same value before and after an assignment, it is an invariant of the assignment. Let P(u, v) be an expression involving variables u and v. P(u, v)[u, v:= el, e2] is obtained from P(u, v) by replacing u by el and v by e2 simultaneously. P(u, v) is an invariant of assignment u, v := el, e2 if

**P(u,v) [u,v := el, e2] = P(u,v)**

**Example 8.3.** Show that p - c is an invariant of the assignment

**p, c := p + 1, c + 1**

Let P(p, c) = p - c. Then

**P (p, c) [p, c := p + 1, c + l]**

**= p — c [p, c := p + 1, c + l]**

**= (p + 1) — (c + 1)**

**= p — c**

**= P(P , c)**

Since (p - c)[p, c := p+l, c+l] = p - c, p - c is an invariant of the assignment p, c := p + 1, c + 1.

**Example 8.4.** Consider two variables m and n under the assignment

**m, n := m + 3, n - 1**

Is the expression m + 3n an invariant?

Let P(m, n) = m + 3n. Then

**P(m, n) [m, n := m + 3, n — l]**

**= m + 3n [m, n := m + 3, n — l]**

**= (m + 3) + 3(n — l)**

**= m + 3 + 3n — 3**

**= m + 3n**

**= P(m, n)**

Since (m + 3n) [ m, n : = m + 3, n - 1] = m + 3n, m + 3n is an invariant of the assignment m, n := m + 3, n - l.

## 8.2 Loop invariant

In a loop, if L is an invariant of the loop body B, then L is known as a loop invariant.

**while C**

**-- L**

**B**

**-- L**



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